

Geiger and Marsden scattering data

Data from Geiger and Marsden's 1913 paper "The Laws of Deflexion of α Particles through Large Angles."

θ	Silver	Gold
150	22.2	33.1
135	27.4	43.0
120	33.0	51.9
105	47.3	69.5
75	136	211
60	320	477
45	989	1435
37.5	1760	3300
30	5260	7800
22.5	20300	27300
15	105400	132000

The predicted values are

$$\hat{y} = \frac{d\sigma}{d\Omega} = \frac{\beta}{(1 - \cos \theta)^2}$$

By fitting β values using ordinary least squares we obtain

	Silver	Gold
β	122	153
R^2	0.999	0.999

where R^2 is the coefficient of determination

$$R^2 = 1 - \frac{\sum (y - \hat{y})^2}{(y - \bar{y})^2}$$

Eigenmath script